

Product Requirements Document

Constituency-Level Social Media Sentiment & Influence Intelligence Platform

Version: 1.0 (Pivoted from civic-issue framing to social-media-native model) **Date:** August 2026

1. What This Product Actually Is

Be precise about the category, because the category determines the legal and ethical obligations that follow.

This is a **social-media sentiment and influence-tracking platform**, scoped to: - Public content from named public accounts (creators, commentators, “new media” news pages) — not private individuals - Aggregated, constituency-level sentiment — not individual-level profiling - Prioritized coverage of **electorally competitive (swing) constituencies**, identified from historical election margins - Prioritized coverage of **high social-media-penetration demographics** within those constituencies

This is **political and media intelligence infrastructure**. It is adjacent to, but distinct from, a generic “civic issue dashboard” — the swing-constituency and named-influencer targeting are inherently electoral-strategy concepts, and the PRD should not obscure that. Calling it purely “policy-making support” software when its targeting logic is built around swing seats will not hold up under scrutiny from a client’s legal team, an election authority, or a journalist, and more importantly, mischaracterizing it internally makes it easy to build the wrong safeguards.

What legitimately distinguishes it from “mass surveillance”: - It doesn’t identify or profile individual private citizens - It doesn’t build who-knows-whom social graphs - It works from public posts by public accounts (already broadcasting to public audiences), not private communications - Output is aggregate sentiment/issue trend data, not dossiers on people

What it is not exempt from just because it’s “public data”: - Platform Terms of Service (crawling/automated collection of named accounts’ content is regulated by each platform’s API terms, regardless of whether the account is public) - India’s DPDP Act 2023 obligations wherever any personal data (including commenters’ public posts) is processed - Election Commission of India’s Model Code of Conduct and rules on political advertising/data use, if the client is a party, candidate, or campaign vendor - Defamation/reputational risk in how sentiment about named commentators is stored, scored, and reported

2. Product Goal

Give a government body, party, or political consultancy a constituency-level read on: 1. What issues and narratives are gaining traction on **social/new media** specifically (as distinct from traditional media) 2.

Which named public accounts (influencers, “new media” news channels) are driving that traction 3. Where this matters most — i.e., in constituencies that are (a) electorally competitive and (b) socially-media-saturated enough that online sentiment is a meaningful proxy for public mood

3. Scope

Dimension	In Scope	Out of Scope
Accounts monitored	Named public pages/creators/news channels with public posts (e.g., political commentary channels, regional “new media” news pages)	Private individual profiles, closed groups, DMs
Platforms	Instagram, Facebook, YouTube, X — via official APIs or licensed data partners	WhatsApp (encrypted, inaccessible), platform scraping that violates ToS
Geography	Constituency-level aggregation, prioritized by swing classification	Booth-level or individual-level mapping
Targeting logic	Swing-constituency prioritization (from historical election margin data) + social-media-penetration filtering	Voter list building, individual voter scoring
Output	Trend reports, narrative/issue attribution to specific accounts, sentiment-over-time, comparative “who’s shaping this narrative” views	Personal dossiers, psychographic profiles of private users

4. Constituency Prioritization Model

4.1 Swing Constituency Identification

Input: historical election results (ECI public data) across at least 2-3 election cycles.

Compute per constituency: - **Margin volatility**: swing in winning margin (%) across cycles - **Vote-share flip frequency**: number of times the winning party/alliance changed - **Closeness index**: margin as % of total votes in the most recent cycle

Classify constituencies into tiers: Safe / Lean / Swing / Toss-up. Product prioritizes Swing and Toss-up tiers for depth of coverage.

4.2 Social Media Penetration Filter

Input: TRAI/telecom internet penetration data, smartphone adoption estimates, regional social media usage surveys, urban/semi-urban/rural classification.

Constituencies are scored on a **Digital Engagement Index** — the degree to which the demographic actively interacts with social platforms (vs. relying primarily on traditional/print/TV media). Only constituencies clearing a minimum threshold get full social-listening coverage; others may still get lighter monitoring but sentiment there is flagged as lower-confidence (traditional media likely dominates).

4.3 Coverage Tiering

- **Tier 1** (Swing/Toss-up + High Digital Engagement): full real-time monitoring, dedicated influencer tracking, weekly reports
 - **Tier 2** (Swing but lower digital engagement, or Safe/Lean but high engagement): monthly monitoring
 - **Tier 3** (Safe + low engagement): quarterly baseline only
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5. Named Account & Channel Monitoring

5.1 Account Universe

Two categories to track, each with a distinct pipeline:

A. Established influencers/commentators — named public accounts with political or civic commentary reach (e.g., large YouTube/Instagram political commentary channels). Monitored for: content topics, framing/sentiment toward government and policy, audience engagement (likes/comments/shares as public metrics), and estimated regional reach (where geographic breakdown is available via platform insights or inferred from language/dialect and content targeting).

B. Rising new-media news channels — emerging pages/channels (regional or national) growing rapidly in reach, distinguished from legacy/traditional media by platform-native format and distribution. Detection method: track follower/subscriber growth rate, view-velocity, and cross-platform amplification (a post spreading from one platform to others) to catch channels *before* they become mainstream-scale.

5.2 What Is Captured Per Account

- Public post content, publish time, platform
- Public engagement metrics (likes, comments count, shares — aggregate counts only, not commenter identities beyond what's needed for sentiment scoring)
- Inferred primary constituency/region of relevance (from content, stated location, or audience-location aggregates provided by platform APIs — never by individual-level geolocation)
- Sentiment/stance classification of the *content itself* toward specific policies or issues

5.3 Explicit Exclusions

- No storage of individual commenter names, profile links, or identifiers
- No cross-referencing of commenters against voter rolls or any other PII dataset
- No attempt to de-anonymize or identify persons behind pseudonymous accounts

6. System Architecture

LAYER 1 – INGESTION

- Official platform APIs (Meta Graph API, YouTube Data API, X API) |
 - Licensed third-party social data providers (for scale/coverage gaps in official APIs – contractually compliant, not scraped)
 - Input: curated account list (Tier 1/2/3 per Sec 4.3) + keyword/hashtag watchlists per constituency
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LAYER 2 – PROCESSING

- Dedup, language ID, Hinglish/regional-language normalization
 - PII scrubbing on any comment-level text retained for context
 - Content classification: civic/political vs. non-relevant
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LAYER 3 – ANALYSIS ENGINE

- Topic/issue extraction (LLM-based)
 - Sentiment & stance classification (multilingual, sarcasm-aware)
 - Account-level narrative attribution ("who is driving this story") |
 - Growth/anomaly detection (rising channels, sentiment spikes)
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LAYER 4 – CONSTITUENCY INTELLIGENCE GRAPH

- Nodes: Constituency, Issue, Account/Channel, TimeWindow, Policy
 - Edges: DISCUSSED_IN, DRIVEN_BY, SENTIMENT_TREND, ADDRESSES
 - Swing-tier and Digital-Engagement-tier as constituency attributes |
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LAYER 5 – REPORTING & ALERTING

- Constituency dashboards (swing-tier prioritized)
 - Influencer/channel leaderboard per constituency/issue
 - Rising-channel early-warning alerts
 - Weekly/monthly narrative briefs
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7. Compliance & Risk — Read This Section Before Building

Area	Requirement	Note
Platform ToS	Use official APIs (Meta Graph API for public Pages, YouTube Data API, X API) or a licensed data vendor with platform agreements in place	Automated scraping of public profiles, even when technically accessible, generally violates platform ToS and can trigger legal action independent of privacy law
DPDP Act 2023	Even “public” comment text can constitute personal data if identifiers survive; PII scrubbing at ingestion is not optional	Get a legal opinion specific to your data-processor/fiduciary structure before commercial deployment
Election law	If any client is a political party, candidate, or campaign vendor, this platform’s outputs (swing-constituency targeting, narrative-driver identification) fall within the ambit of ECI’s Model Code of Conduct and political advertising/data-use rules during election periods	Build a compliance gate that restricts certain features (e.g., real-time influencer targeting) during the MCC window
Defamation exposure	Sentiment scoring and “narrative driver” attribution about named individuals (commentators, channels) is reputationally and legally sensitive if published or acted on incorrectly	Keep account-level sentiment analysis internal-use-only; do not publish scored rankings of named individuals externally
Data retention	Define retention limits for raw post/comment text vs. aggregated scores	Aggregate insights can be retained longer than raw scraped text

8. Output Examples

Constituency Brief (Tier 1, Swing): > **Constituency:** [X], **Tier:** **Swing / High Digital Engagement** > Dominant narrative this week: unemployment scheme delays. Sentiment: -0.61 (down from -0.34 last week). > Primary drivers: a regional Instagram news page (340K

followers, +18% growth this month) and a political commentary YouTube channel whose latest video (2.1M views) framed the issue around delayed disbursement. > Cross-platform spread detected: content first appeared on Instagram, amplified to X within 6 hours. > Recommended: policy communication addressing disbursement timeline, given this is displacing a previously dominant narrative (infrastructure) in a swing seat.

Rising Channel Alert: > New regional YouTube news channel crossed 500K subscribers in 45 days (growth rate in top 2% of tracked channels). Content skews toward [constituency cluster]. Recommend adding to Tier 1 monitoring.

9. Technology Stack

Layer	Technology
Ingestion	Official platform APIs + licensed data partners; Airflow for scheduling
Streaming/processing	Kafka + Flink, or managed equivalents
Storage	PostgreSQL (metadata), Neo4j (constituency-issue-account graph), Elasticsearch (text search)
ML/NLP	Multilingual transformer models (IndicBERT-class) for sentiment; LLM-based topic/narrative extraction
Dashboard	React/Next.js, Mapbox for constituency visualization
Infra	India-region cloud (data residency for DPDP compliance)

10. Open Decisions Before Build

- Legal review:** engage counsel on DPDP + election law applicability *before* finalizing which accounts/constituencies are monitored, since this determines architecture (retention windows, MCC feature-gating).
- Data sourcing:** confirm which platforms’ official APIs actually expose the account-level and geo-aggregate data this design assumes (Meta and YouTube data availability differs significantly from X, and coverage gaps may require licensed vendors).
- Client type:** government ministry vs. political party vs. consultancy changes the compliance obligations substantially — this should be locked before go-to-market.